

Raw data

dimensions of macro indicators and how to move between them

WHAT A NUMBER MEANS | WHAT DERIVES FROM WHAT | PRACTICE

exit10.ru

github.com/MikeMike13/MACRO_RAW_DATA

Four numbers in the news — four different dimensions

before computing anything further, we need to know what exactly was published

> *"Deflation in Russia in August was 0.08%, annual inflation accelerated to 6.33%"*

— Interfax, 11 Sep 2026

`growth vs previous month (m/m) and vs a year ago (y/y)`

> *"Federal budget revenue for January–August rose 9.2% year on year"*

— Expert, 9 Sep 2026

`cumulative since January (cmlt, cmlt/cmlt)`

> *"Industrial production in Russia fell 0.6% in August year on year"*

— Reuters / The Moscow Times, 23 Sep 2026

`y/y only — no level and no m/m`

> *"Russia's international reserves as of 1 August amounted to 720.348 bn dollars"*

— Rossiyskaya Gazeta, 7 Aug 2026

`a stock at a date, in dollars, in billions`

media quotes, August–September 2026; translated from Russian

Seven questions to ask of any number

any value can be described by these characteristics — they decide what can be done with it

field	question	example from the news
`metric`	a number "as is", a share, a rate or already % of GDP?	unemployment 2.2% — a share, not a level
`transf`	a level or growth? growth vs what?	−0.08% — vs the previous month
`stock_flow`	over a period or at a date?	reserves — "as of 1 August", a stock
`dim · currency`	in what units?	RUB tn, USD bn, m³
`mult`	in what scale is it stored?	GDP in billions, exports in millions
`real_nominal`	current or constant prices?	GDP 56.2 tn — nominal, +1.3% — real growth
`freq`	day, week, month, quarter?	weekly inflation ≠ monthly

I

What a number means

units, stocks and flows, percent and percentage points

One folder, three storage scales: without mult the error is a thousandfold

stored value × mult = the quantity in roubles, dollars, cubic metres

indicator	stored value	mult	actual quantity
Federal budget revenue, year-to-date	25,928,748,210,643	1	25.9 tn RUB
Nominal GDP, quarter	56,166.3	10^9	56.2 tn RUB
Exports of goods, quarter	100,692.5	10^6	100.7 bn USD
International reserves	720.3	10^9	720.3 bn USD
Gas production, month	39,578.1	10^6	39.6 bn m^3

Before dividing one series by another (e.g. by GDP), convert both to absolute units. Choose the display scale only after the calculation.

Flows are summed, stocks taken at the end, prices averaged

this decides aggregation, the FX rate to use and the ratio to GDP

	flow	stock	average avg
what it is	how much happened over the period	how much there is at a date	average level over the period
examples	GDP, budget revenue, exports	reserves, M2, debt	prices, FX rate, interest rate
month → quarter	sum of three months	end value	average of three months
rolling window	12-month sum	average stock	average
to another currency	at the average rate	at the rate on the date	at the average rate
to GDP	quarter/quarter or 4Q/4Q	to 4-quarter GDP	usually not

“As of 1 August” means end of July

a stock is dated at the end of the period: USD 720.3 bn "as of 1 August" is the July 2026 value



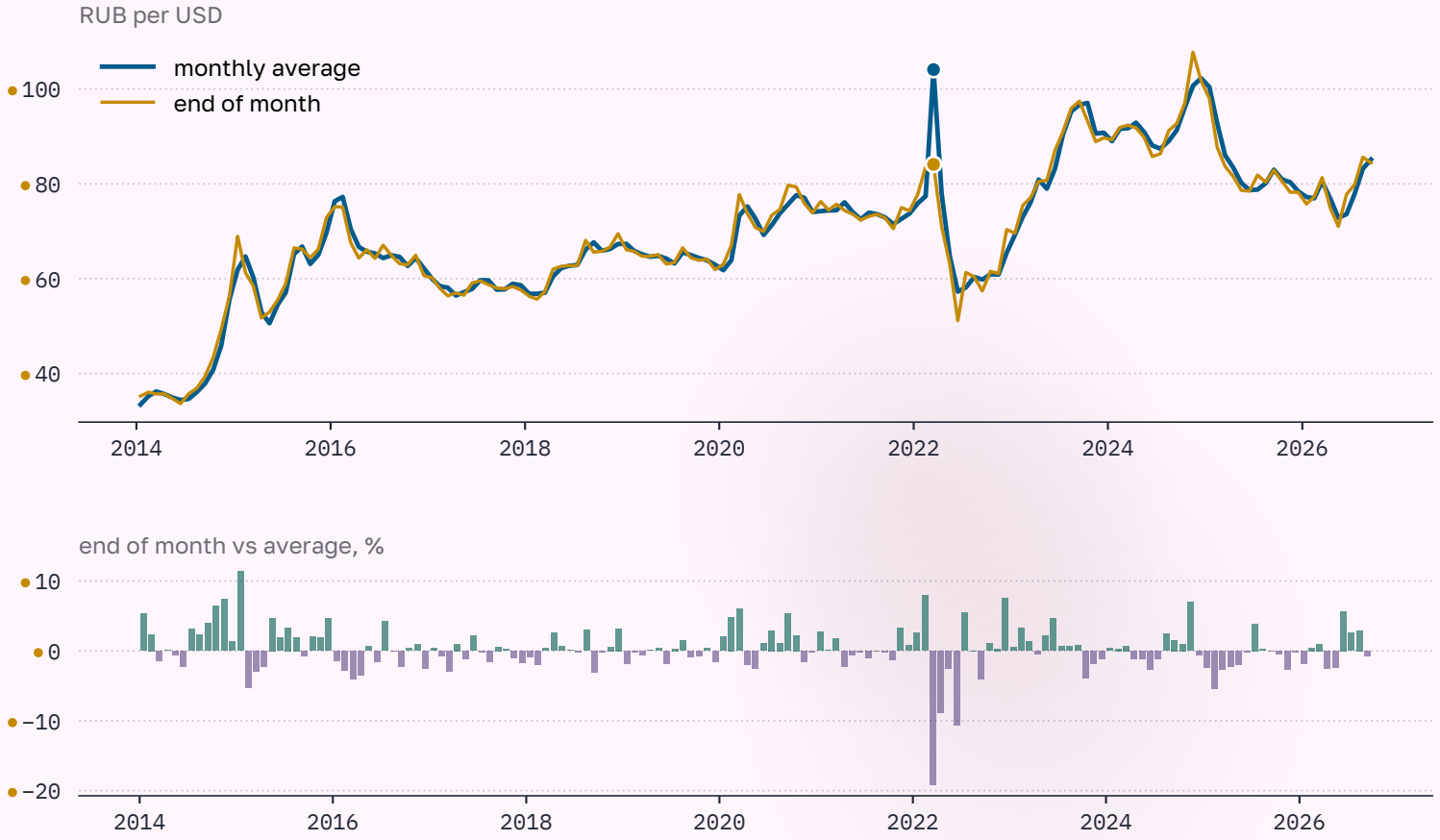
> *“Russia's international reserves as of 1 August amounted to 720.348 bn dollars”*
>
> — Rossiyskaya Gazeta

Agencies say "as of the 1st"; in the series the same value is the end of the previous month. A quarter = the value in March, June, September, December.

data: Bank of Russia; quote: Rossiyskaya Gazeta, 7 Aug 2026; translated from Russian

End-of-month or average rate: answers differ by up to 19%

RUB per USD; a flow is converted at the average rate, a stock at the rate on the date (characteristic eop_avg)



THE RATE MATTERS

March 2022: average rate 104.1, end of month 84.1 RUB/USD

	in roubles	rate
USD 1 bn of exports in March (flow)	RUB 104 bn	average
USD 1 bn of reserves on 31 March (stock)	RUB 84 bn	end-month

The same dollar — a 19% gap purely from the choice of rate. That is why the characteristics say what the series is: a flow, a stock at a date or a period average.

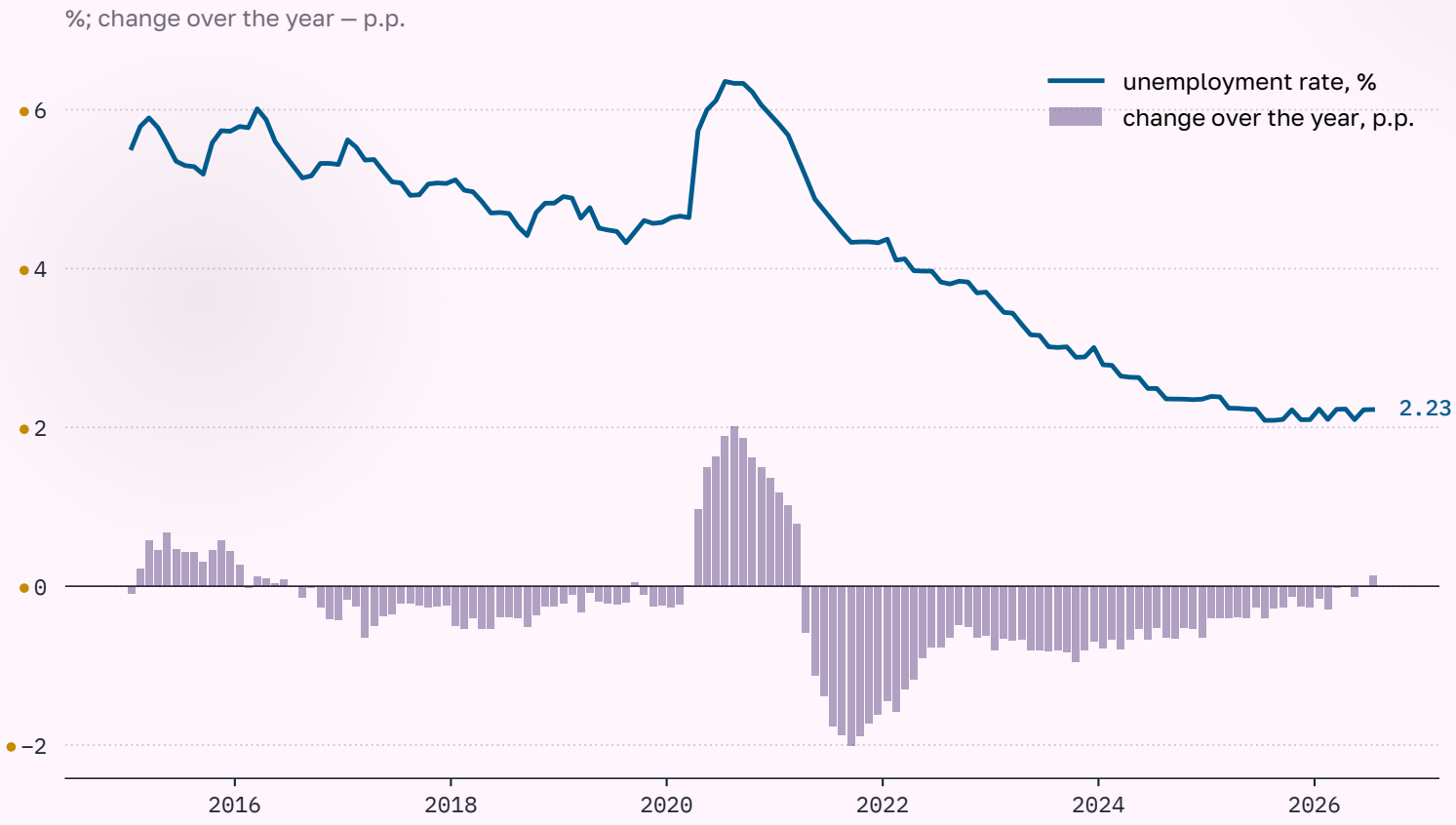
One dynamic, six ways to publish it

L — level, C — cumulative since January, k — periods per year

transf	formula	how it sounds in the news
none	L(t)	RUB 3.8 tn collected in August
mom · qoq	L(t) / L(t−1) − 1	prices fell 0.08% in August
wow	L(t) / L(t−1 wk) − 1	prices rose 0.06% over the week
yoy	L(t) / L(t−k) − 1	output 0.6% lower than a year ago
ytd	L(t) / L(December) − 1	prices up 4.67% since the start of the year
cmlt	C(t) = Σ L since January	January–August: RUB 25.9 tn
cmlt_cmlt	C(t) / C(t−k) − 1	January–August vs January–August: +9.2%

A share changes in percentage points, not in percent

unemployment 2.09% → 2.23%: +0.14 p.p., not "by 6.5%"



For a level the change is a ratio: L(t) / L(t−1) − 1.

For a quantity already in percent (a share, a rate, a growth rate) it is a difference: x(t) − x(t−1).

The acceleration of inflation is measured the same way: "from 5.98% to 6.33% y/y — by 0.35 p.p."

p.p.

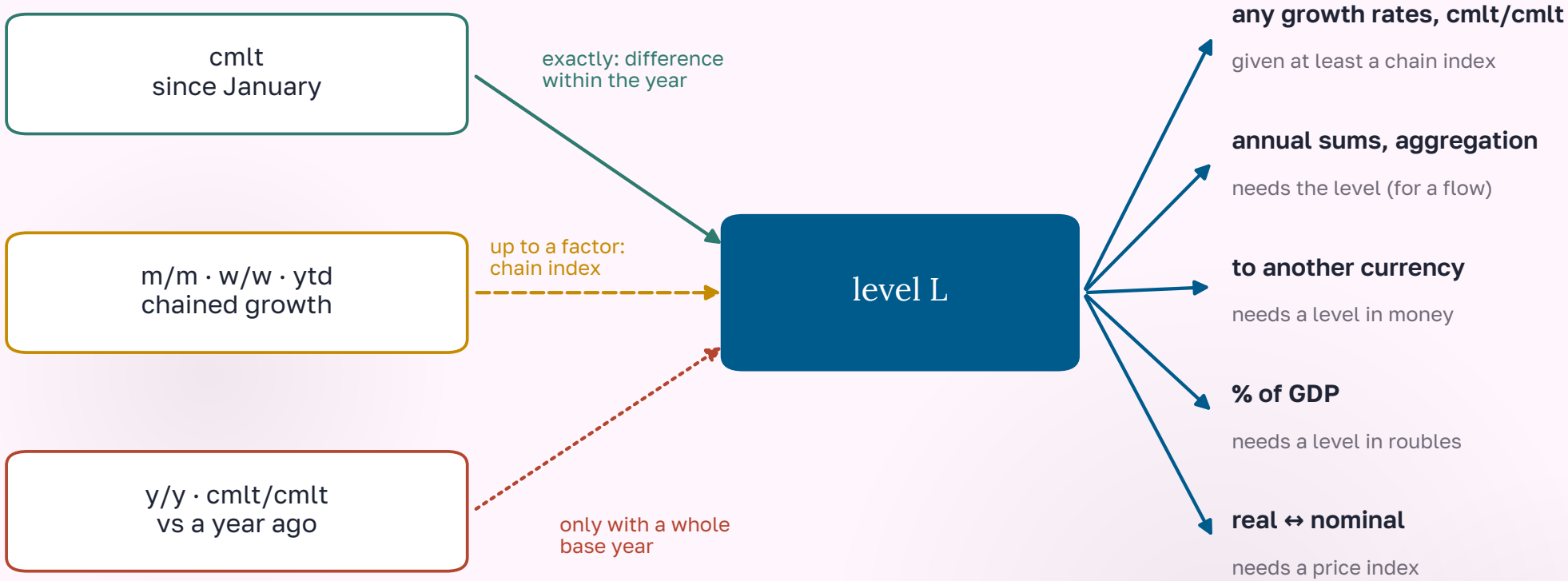
II

What derives from what

everything goes through the level — the only question is whether it can be recovered

Everything goes through the level

what can be derived from a series depends on how precisely its level can be recovered



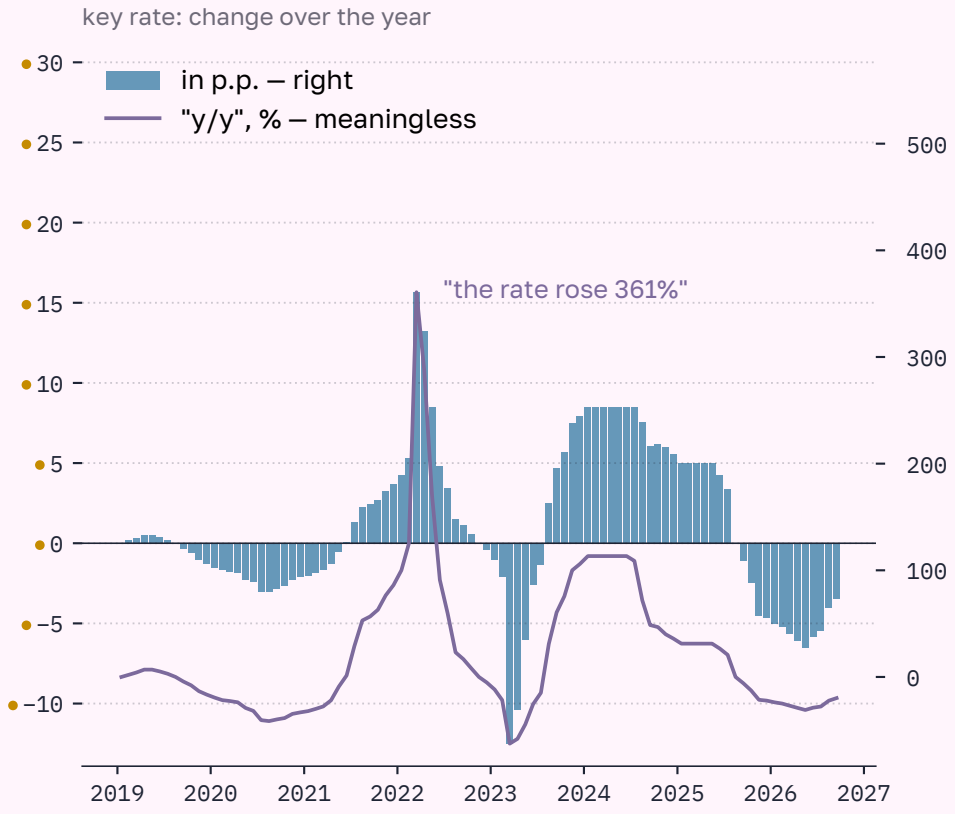
What derives from what: one table

● possible ○ needs extra data (anchor, base year, price index) ◇ computable but meaningless

	level	index	vs prev. period	y/y	vs Dec-ember	year-to-date	cmlt/ cmlt	quarter/ year	rolling window	currency	% of GDP	real/ nominal
money, flow, nominal	●	●	●	●	◇	●	●	●	●	●	●	○
money, stock	●	●	●	●	●	◇	◇	●	●	●	●	○
physical units, flow	●	●	●	●	◇	●	●	●	●	◇	◇	●
year-to-date (cmlt)	●	●	●	●	◇	●	●	●	●	●	●	○
vs previous period (m/m, w/w)	○	●	●	●	●	○	◇	●	●	◇	◇	○
vs end of last year (ytd)	○	●	●	●	●	○	◇	●	●	◇	◇	○
y/y (yoy)	○	○	○	●	○	○	○	○	●	◇	◇	●
cmlt/cmlt	○	○	○	○	○	○	●	●	◇	◇	◇	●
share, %	●	◇	◇	◇	◇	◇	◇	●	●	◇	◇	◇
rate, % p.a.	●	◇	◇	◇	◇	◇	◇	●	●	◇	◇	○

Computable — but meaningless

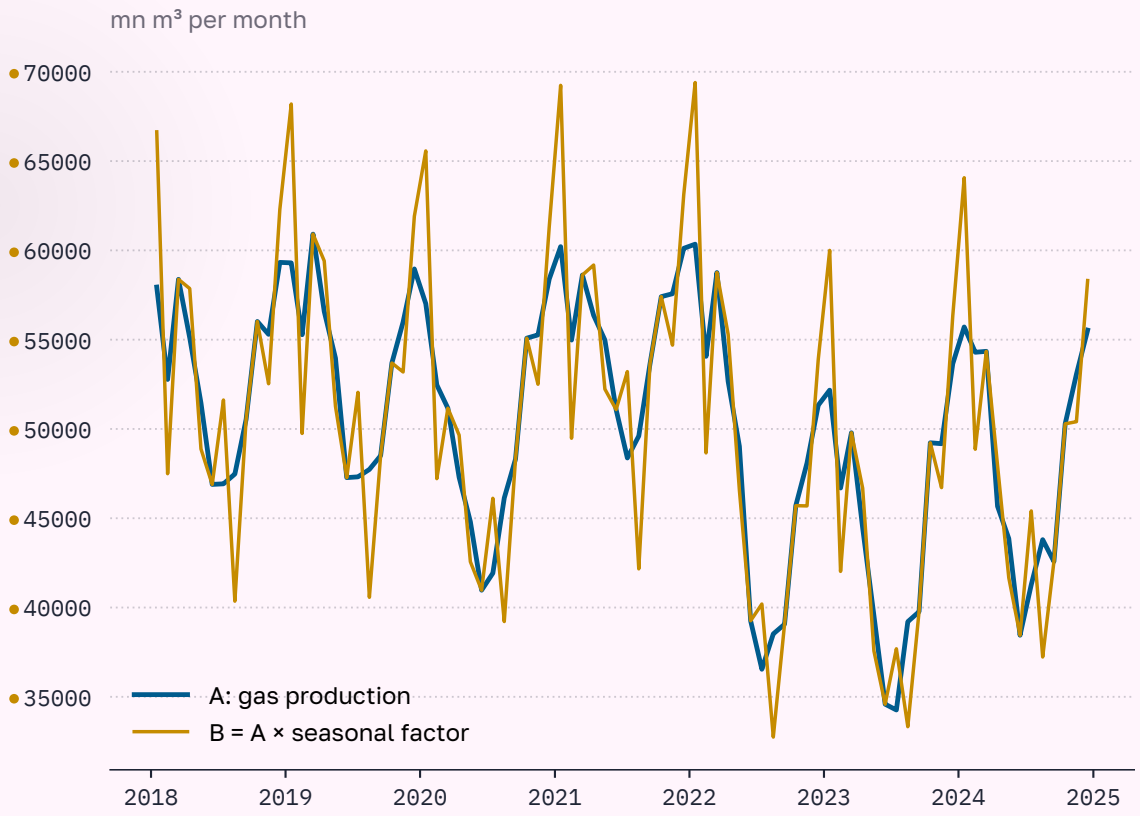
the formula works, but the result means nothing: ◇ in the transformation table

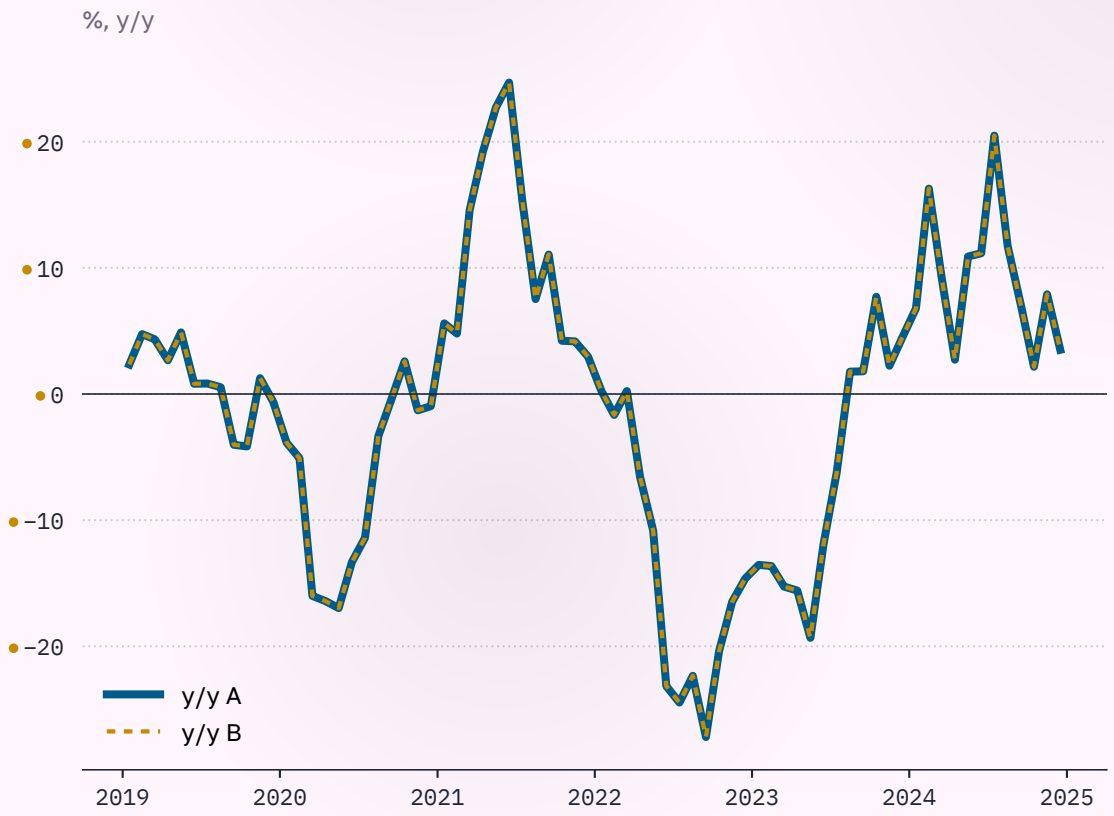


what was computed	result	why it is meaningless
rate "y/y" in %	+361% (2022)	the change is in p.p.
12-month sum of reserves	9.0 tn $	snapshots are not summed
budget revenue vs December	−45% in January	December is seasonal
gas production to GDP	0.26%	m³ vs roubles
m/m inflation in dollars	% / FX rate	growth has no currency
a stock to quarterly GDP	overstated ×4	a stock goes to annual GDP

Two different series — the same y/y

B = A × the same factor for each calendar month: y/y coincide exactly, monthly dynamics do not





PROOF

M/m knows the shape of a series, not its scale

from y/y and cmlt/cmlt the level is recovered only given one whole known year

	what we do	what we get	limitation
series A and 3A	m/m, y/y, cmlt/cmlt coincide	annual volume differs threefold	needs an anchor — one level
m/m → chain index	Π(1 + x) — base 100 is arbitrary	enough for any growth rates	not for % of GDP or FX
y/y + year 2018	L(t) = L(t−12)·(1+g)	recovered exactly: yes	impossible without 2018
ytd ↔ m/m	1+m = (1+ytd)/(1+ytd prev.)	the same information	convert exactly

anchor and base year

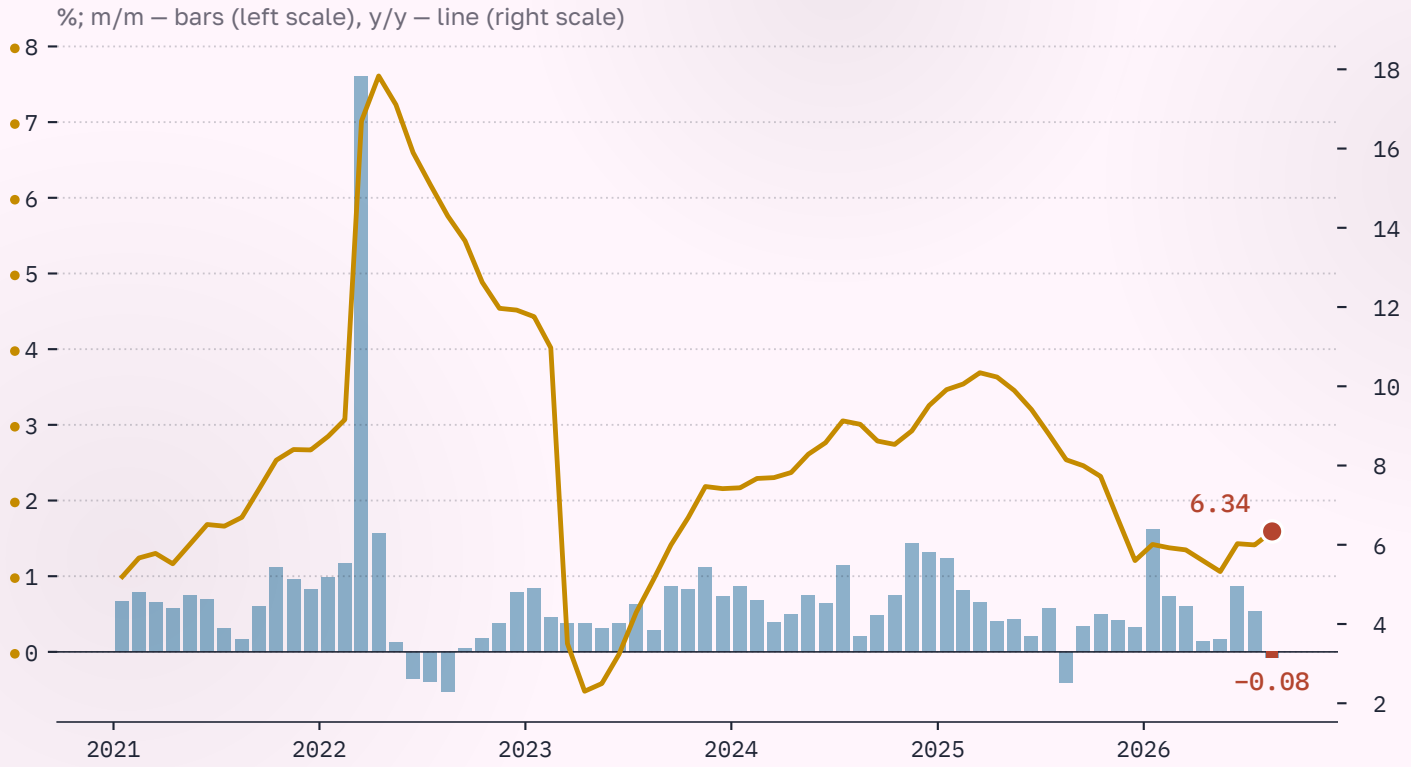
III

Picking up numbers from the news

and computing further; the number from the news is red — in the quote and on the chart

From m/m we build y/y and year-to-date — and match Rosstat

August 2026: y/y 6.34% (Rosstat: 6.33%), year-to-date 4.67% (4.67%)

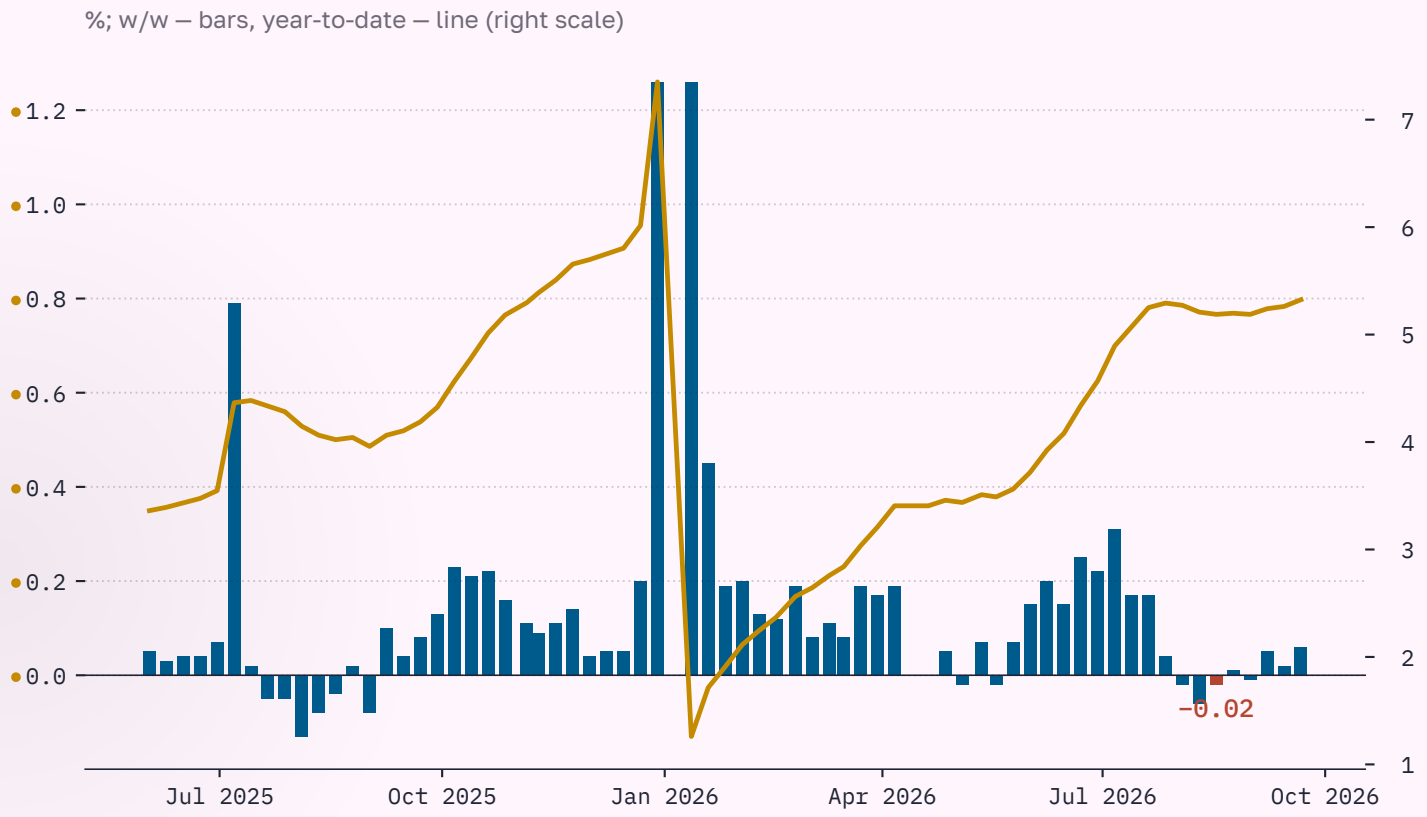


> *"Deflation in Russia in August was 0.08%, annual inflation accelerated to 6.33%"*
>
> — Interfax

A chain index of 224 monthly changes gives 6.34% vs 6.33%: m/m is published rounded to two decimals, and the error accumulates. Year-to-date matches to the hundredth.

Week by week: −0.02% for 11–17 August

price change vs the previous week, %; from it — a chain index and year-to-date growth



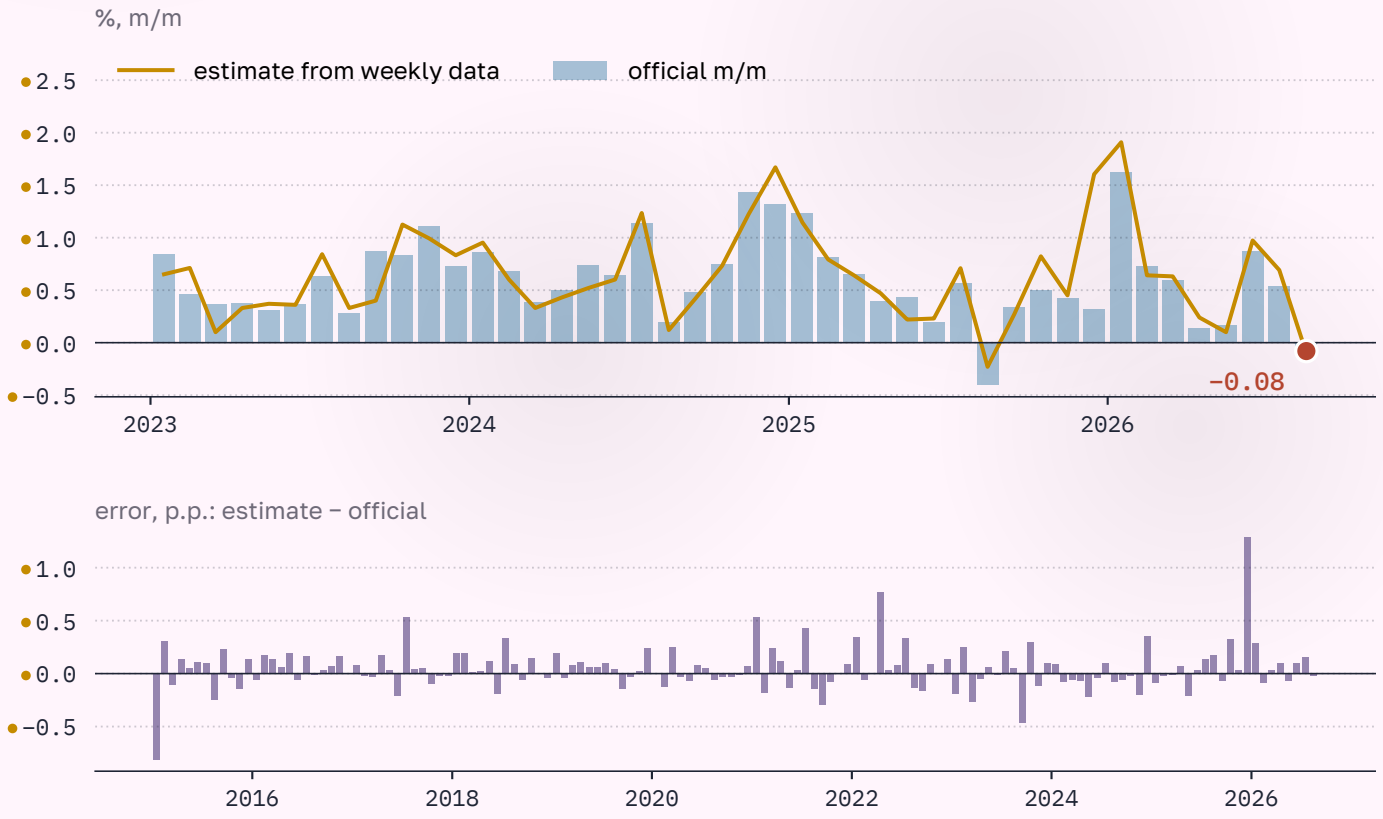
> *"In the week of 11–17 August 2026 prices fell 0.02%"*
>
> — Interfax

The weekly index is the same chained growth as m/m, only more frequent. A 52-week year and weeks straddling two months make a "month" and a "year" from weekly data an approximation — how accurate, see the next slide.

data: Rosstat; quote: Interfax, 19 Aug 2026; translated from Russian

Weekly data guess the month to within about 0.2 p.p.

m/m from the index of the last week of the month vs the official m/m; below — their difference



> *"the weekly dynamics, which also pointed to an August price decline of 0.08%"*
>
> — Alfa Investor

accuracy since 2015

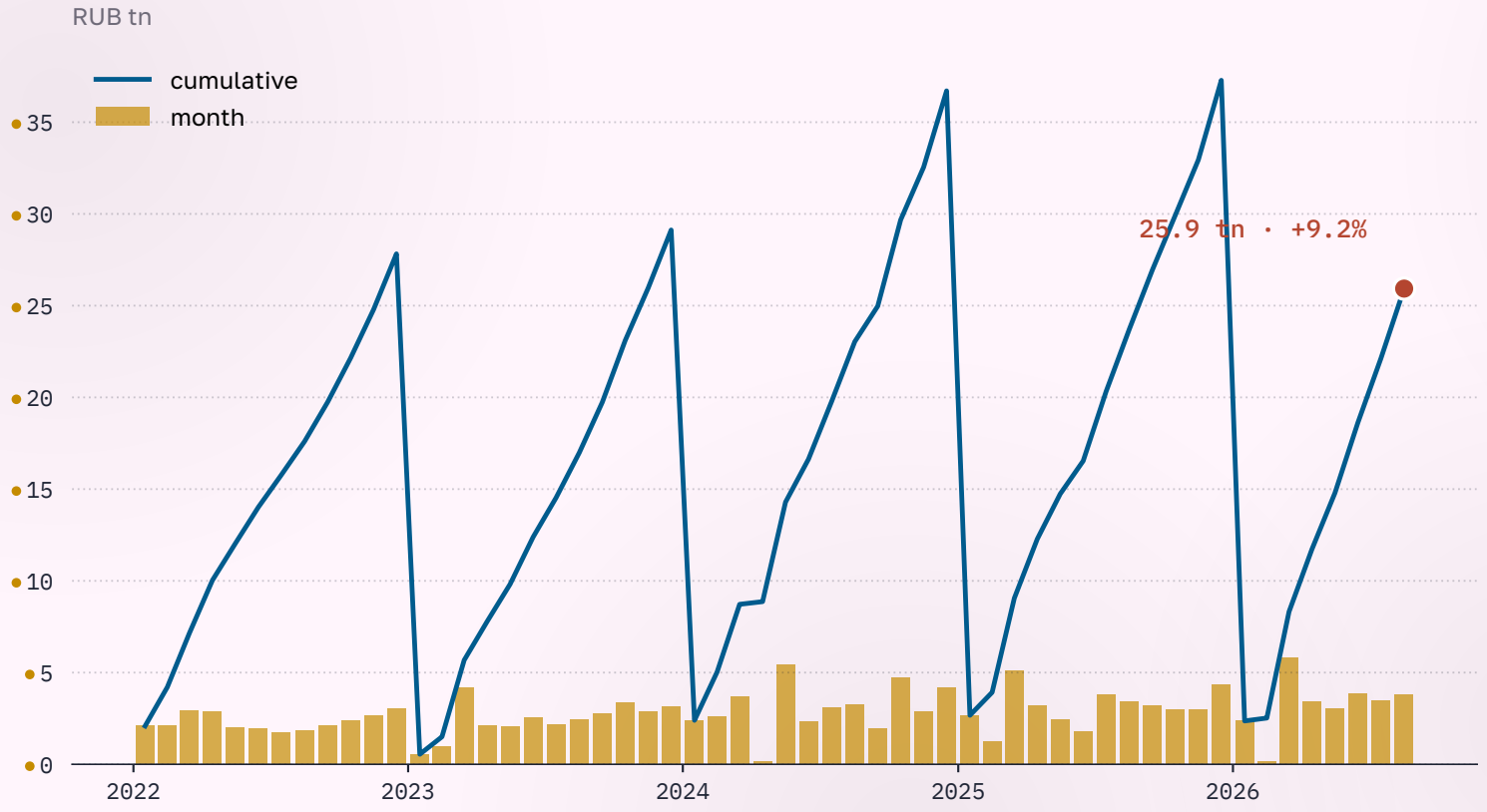
RMSE	0.22 p.p.
median error	0.03 p.p.
bias	+0.05 p.p., p = 0.01
|error| > 0.2 p.p.	20% of months
RMSE / spread of m/m	28%
correlation	0.96

Last-week estimate for August: −0.10%. Direction and size are reliable, the exact number is not.

data: Rosstat; quote: Alfa Investor (Interfax), 11 Sep 2026; translated from Russian

RUB 25.9 tn year-to-date is a sawtooth, not a series

decumulate to months: August = 25.93 − 22.11 = RUB 3.82 tn; January–August vs January–August 2025: +9.2%

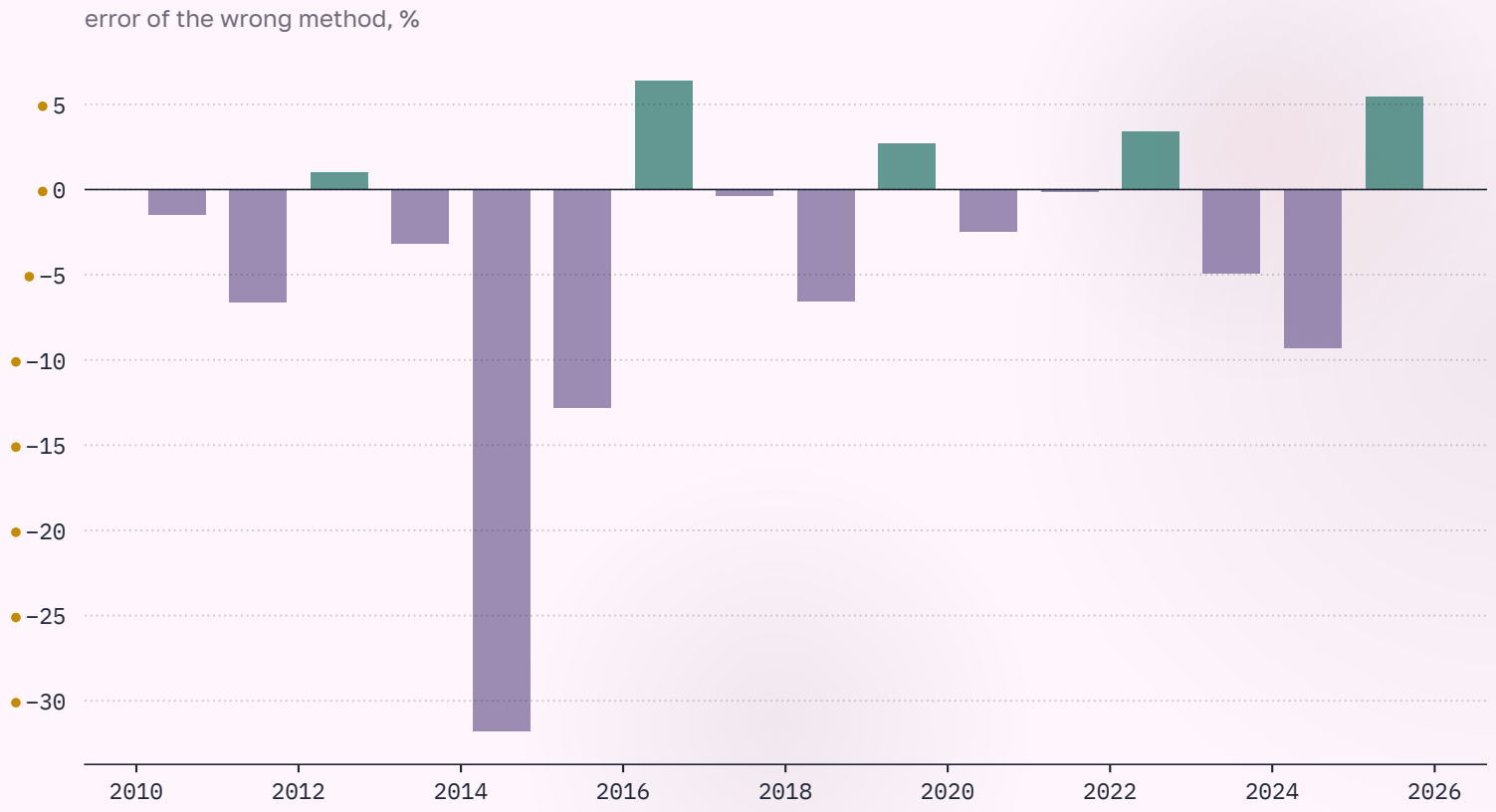


> *"Budget revenue rose 9.2% to 25.9 tn roubles"*
>
> — T–Zh

A month = the difference of neighbouring cumulative values within the year. The annual volume comes directly from cumulative values: C(t) + C(December) − C(t−12). 9.2% is cmlt/cmlt: January–August vs January–August.

A cumulative sum must not be divided by the last month's rate

annual revenue in dollars: "December cumulative / December rate" vs "each month at its own rate"

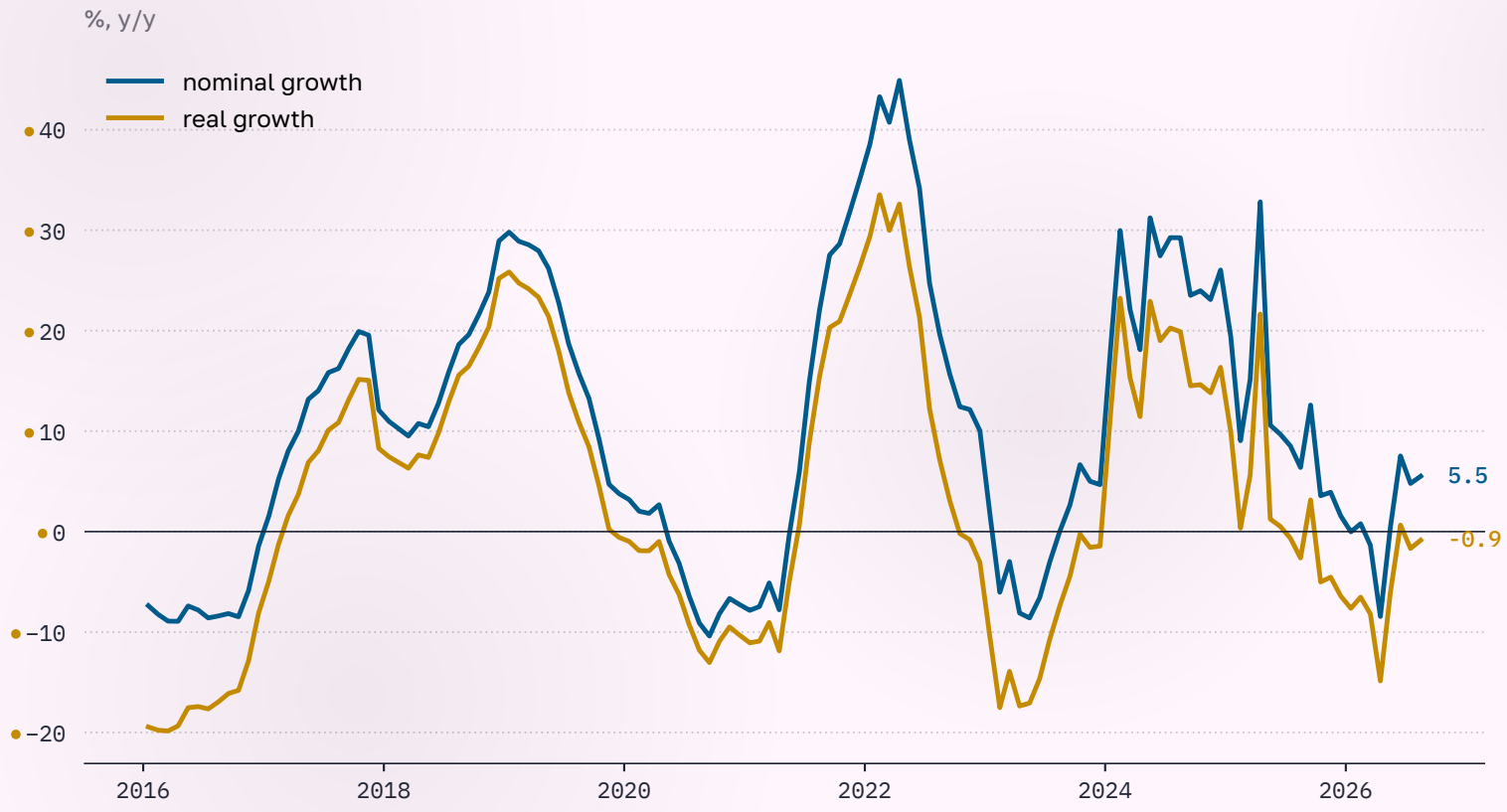


year	right, $ bn	wrong
2022	411	425
2023	338	321
2024	396	359
2025	451	476

January's roubles came in at January's rate. The right way: decumulate to months → convert each month at its average rate → cumulate again.

Revenue +5.5% nominal, −0.9% real

12-month sum, y/y; real — deflated by the CPI, in August 2026 prices



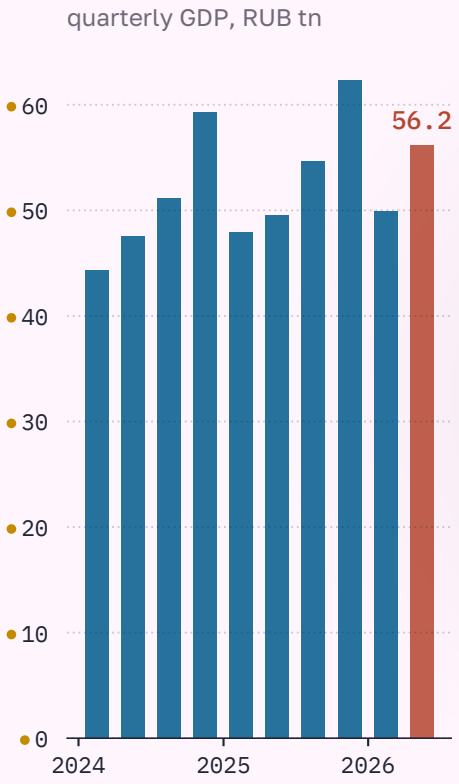
Nominal = volume × prices.

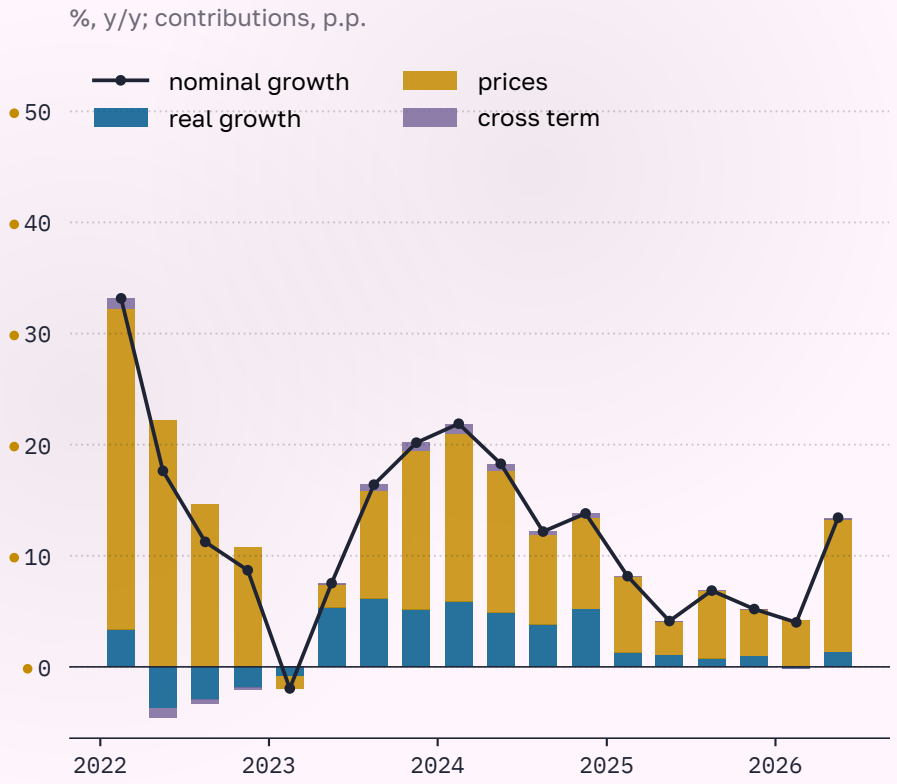
A real series = nominal / a price index. Here the index is the CPI chained from m/m.

Any base ("in which month's prices") can be chosen: the level shifts by one factor, growth rates do not change.

56.2 tn nominal, +1.3% real — the rest is prices

Q2 2026 vs Q2 2025: nominal +13.4% = real growth +1.3% + deflator +12.0% + cross term +0.2%





> *"In current prices GDP amounted to 56.2 tn roubles"*
>
> — Vedomosti

$$(1 + n) = (1 + r)(1 + p)$$

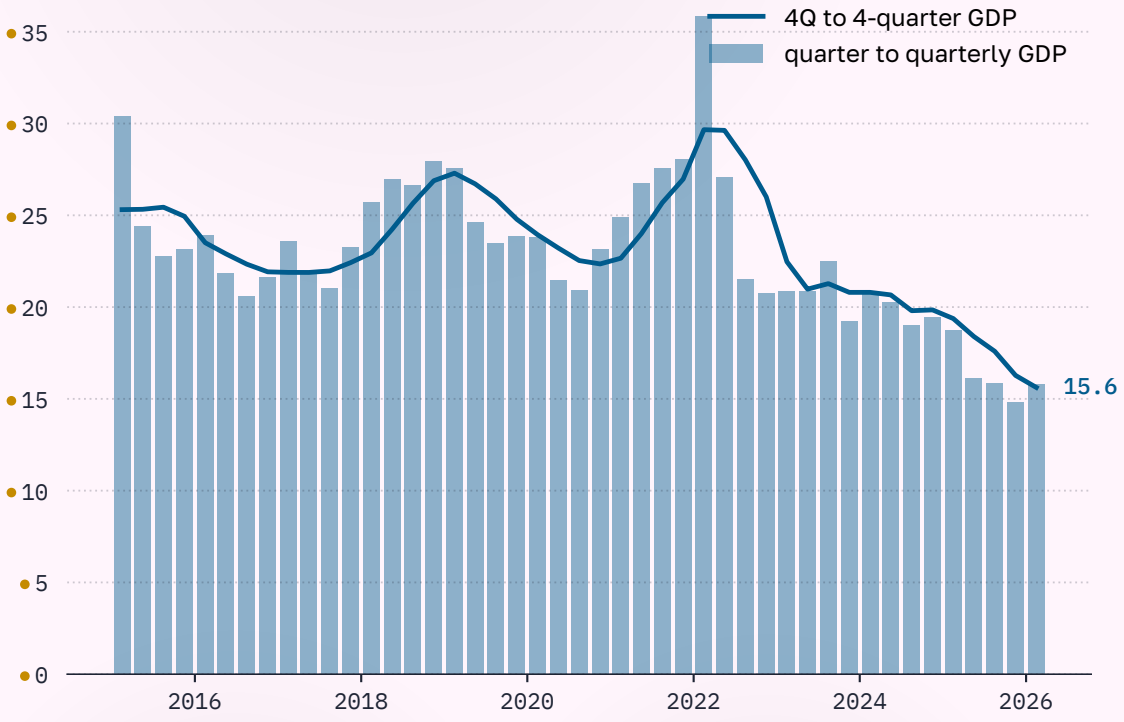
Real growth of 1.3% matches Rosstat.

data: Rosstat; quote: Vedomosti, 11 Sep 2026; translated from Russian

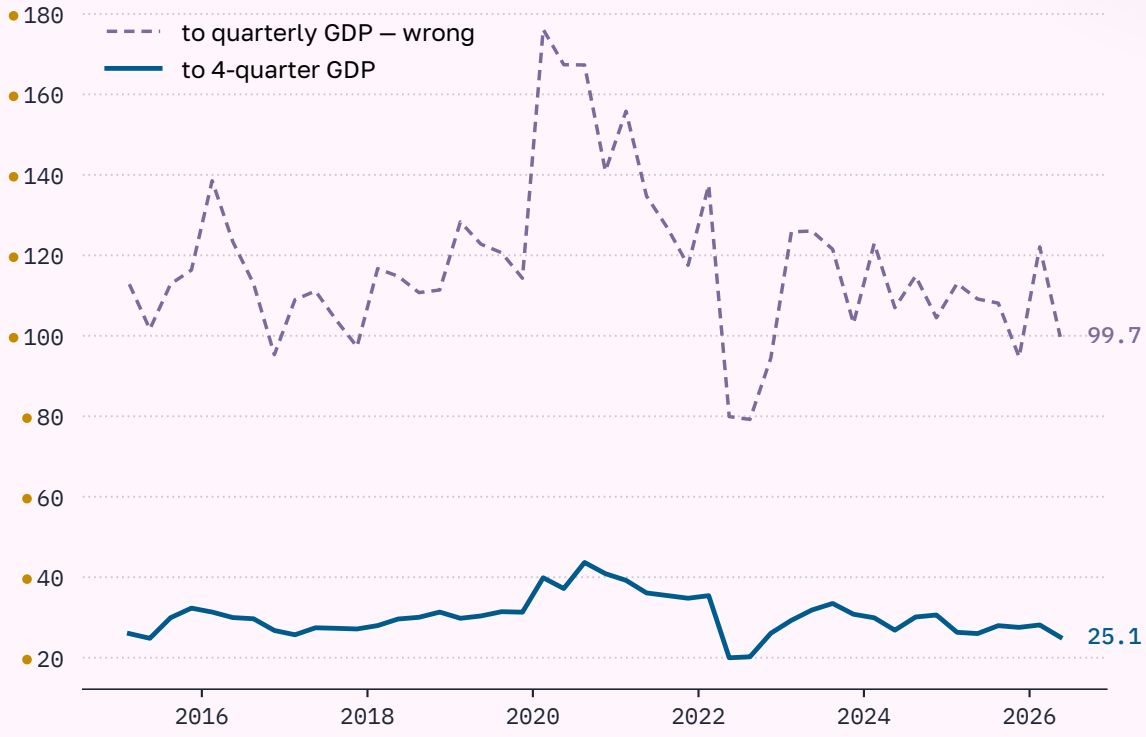
Flows and stocks are related to GDP differently

a flow — quarter to quarter or 4Q to 4Q; a stock — only to 4-quarter GDP

exports of goods, % of GDP



international reserves, % of GDP

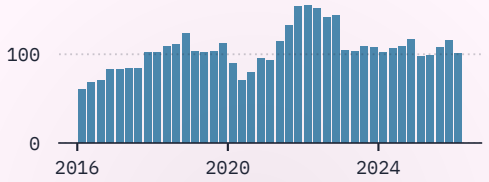


% of GDP

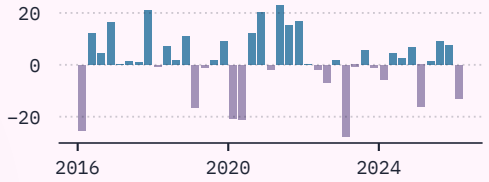
One level, eleven metrics

exports of goods: everything standard is computed from the quarterly level in dollars (tf.all_metrics)

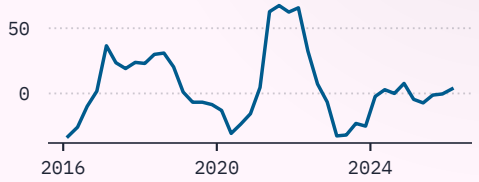
level, $ bn
L(t)



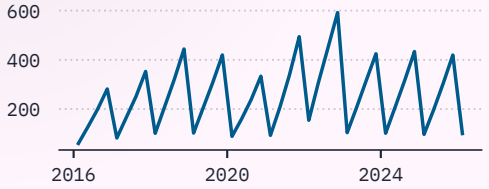
vs prev. quarter, %
L(t)/L(t−1) − 1



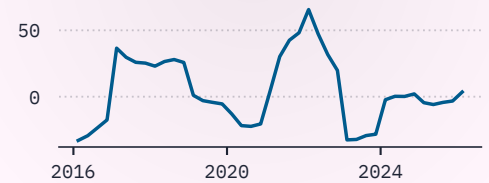
y/y, %
L(t)/L(t−4) − 1



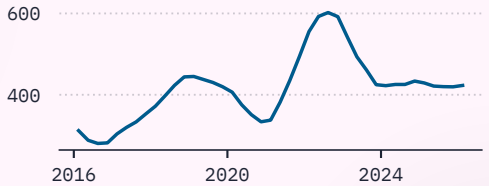
year-to-date, $ bn
C(t) = Σ L since Q1



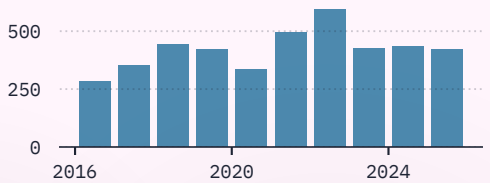
cmlt/cmlt, %
C(t)/C(t−4) − 1



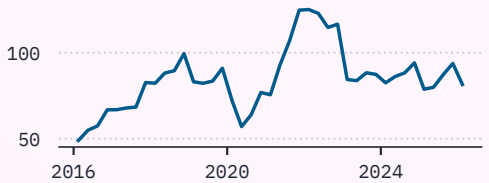
4-quarter sum, $ bn
Σ L(t−3…t)



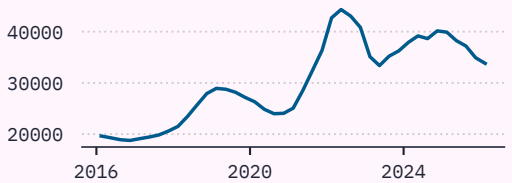
annual, $ bn
Σ L over 4Q



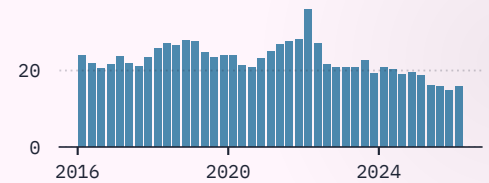
index, 2021 = 100
100·L / L̄(2021)



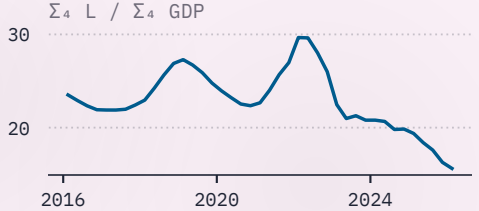
in roubles, 4Q, bn
at the average rate



% of GDP, quarter
L / quarterly GDP

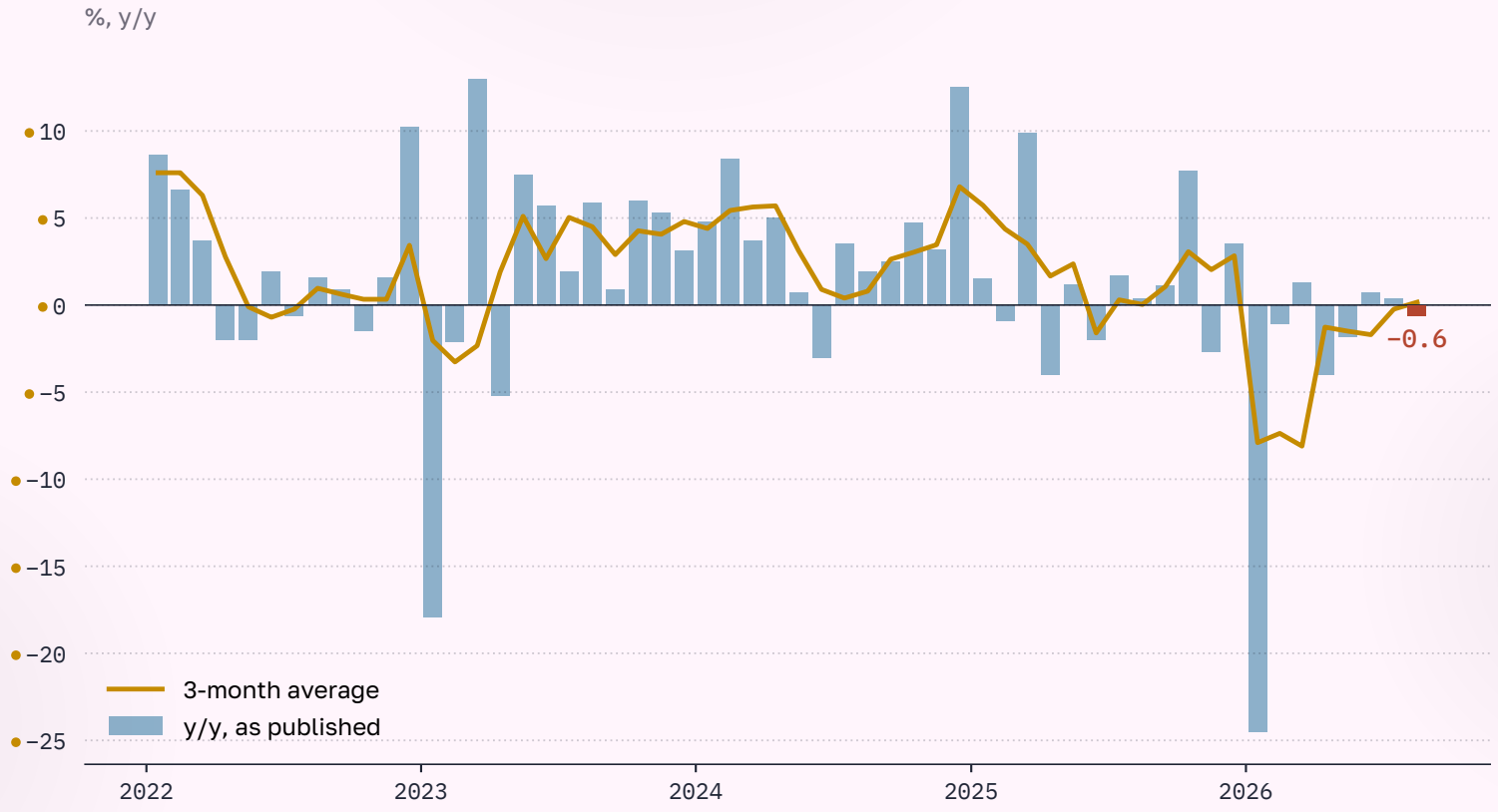


% of GDP, 4Q
Σ₄ L / Σ₄ GDP



Monthly dynamics cannot be extracted from −0.6% y/y

y/y links August only to August; the seasonally adjusted m/m is a separate release and cannot be derived from y/y



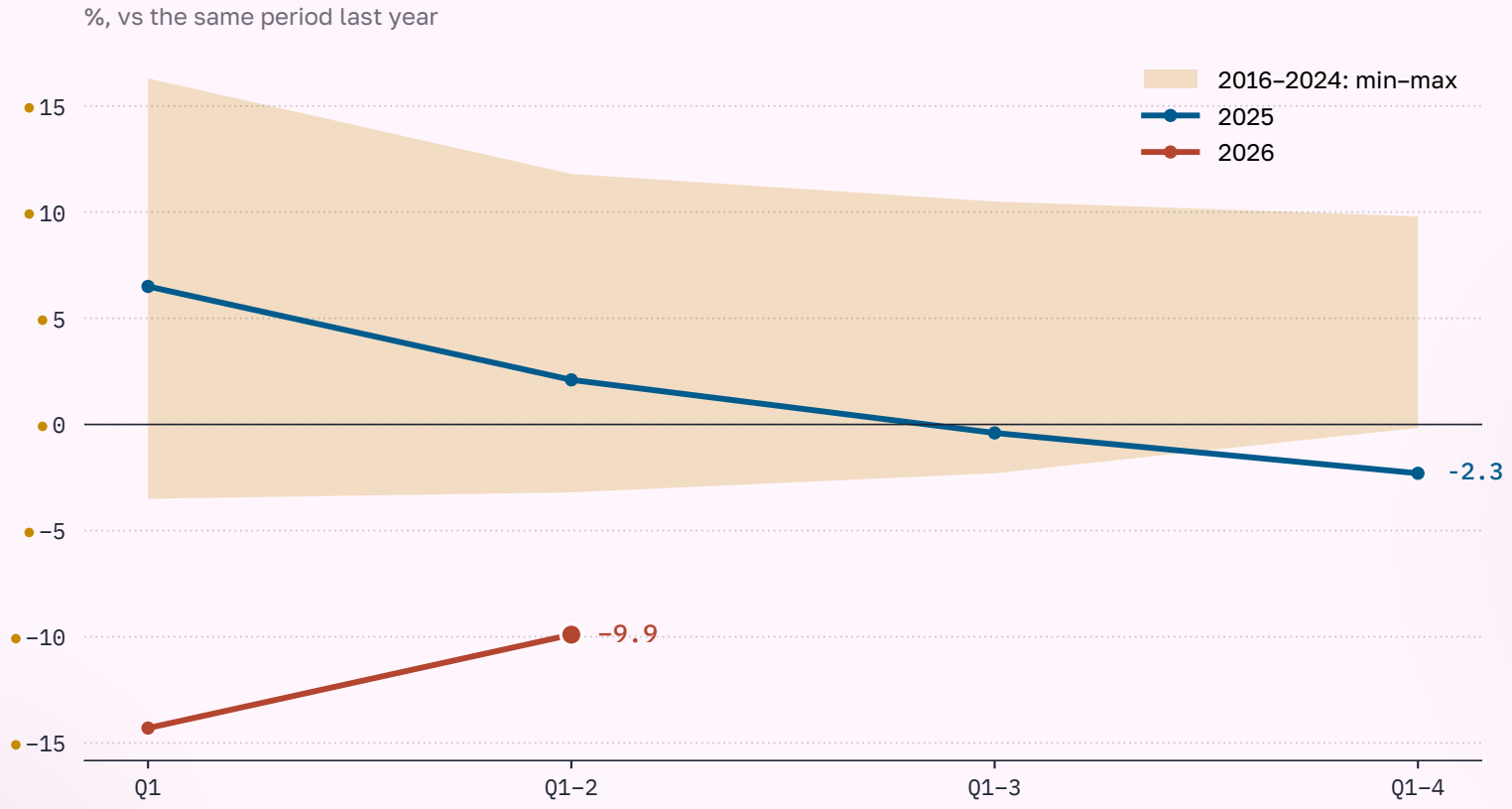
> *“Industrial output in Russia fell 0.6% year on year in August 2026”*
>
> — Finmarket

Y/y can only be smoothed — and only with a trailing window: a centred one at the end of the series relies on months not yet released.

data: Rosstat; quote: Finmarket, 23 Sep 2026; translated from Russian

Investment −9.9% in H1 — the worst start since 2016

cmlt/cmlt by quarter; 2025 and 2026 — lines, 2016–2024 — the range



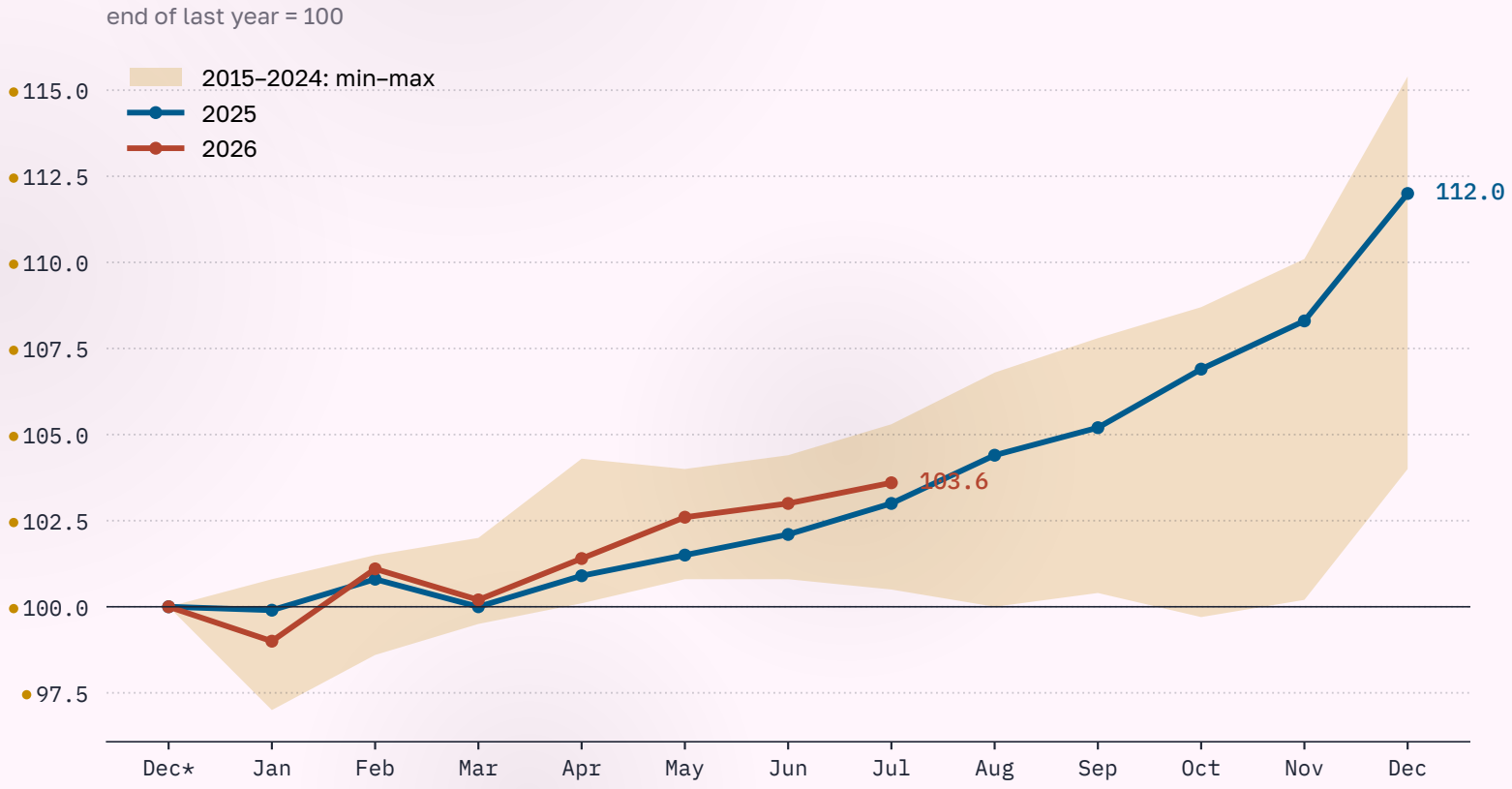
> *"Investment fell 9.9% over the half-year"*
>
> — Mail Finance

Each value includes all earlier quarters of the year, so the line calms down towards year-end. The Q4 value is growth over the year. Quarterly growth cannot be extracted without base-year levels.

data: Rosstat; quote: Mail Finance, September 2026; translated from Russian

Money supply: the path within the year against past years

M2X ex FX revaluation; end of previous year = 100, January = 100·(1 + growth in January), etc.



A ytd series is already measured from December, so the normalised profile is the published 100 + ytd.

For a series of levels the first year would drop out: there is nothing to normalise it by.

The December jump is seasonality, not news: compare with the same month of past years.

Eight typical mistakes

each is a mismatch between the characteristics of a series and the operation applied to it

mistake	the right way
cumulative / last month's rate	decumulate, convert, cumulate again
a stock / one quarter's GDP	a stock — only to 4-quarter GDP
sum of an incomplete quarter	aggregate complete periods only
millions and billions in one ratio	both series × mult before dividing
nominal growth vs real growth	net out prices: (1+n)/(1+p) − 1
annualised non-SA m/m	seasonally adjust first
centred smoothing at the end of a series	last points — trailing only
a change of a share "in percent"	a difference in p.p.

MATERIALS

You can run all of this yourself

notebook, module and data — an open repository

file	purpose
`raw_data_master_pgh.ipynb`	bilingual notebook: theory, transformation map, live examples
`transform.py`	module: characteristics, aggregation, growth, currencies, % of GDP, charts
`RAW_DATA/`	14 series in Excel: Rosstat, Treasury, Bank of Russia
`MANUAL_CHARACTERISTICS`	series characteristics — described by hand in the notebook

The exit10 series: seasonal adjustment → trend extraction → raw data

seasonality	`github.com/MikeMike13/SEASONAL`
trend	`github.com/MikeMike13/TREND`
raw data	`github.com/MikeMike13/MACRO_RAW_DATA`

thank you